

Averting a lost decade in HIV elimination

Continued funding for prevention and treatment is critical for HIV elimination efforts

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Abstract

Funding for HIV prevention and treatment in the United States faces growing uncertainty, and sustained reductions could reverse decades of progress toward HIV elimination. We used INFORM-HIV, an agent-based network model calibrated to HIV prevalence among Black men who have sex with men (MSM) in Chicago, to project how reductions in HIV program funding would affect HIV transmission in this population over the next decade. Reductions in treatment (ART) and prevention (PrEP) funding were projected to produce substantial increases in HIV incidence. A 25% reduction in ART funding increased projected 10-year HIV incidence by approximately [X]%, whereas a 25% reduction in [PrEP / both ART and PrEP] increased incidence by [Y]%. In a setting where HIV incidence is already rising in some populations, these findings underscore the importance of sustained funding for HIV prevention and treatment for the United States to successfully eliminate HIV over the next decade.

Keywords: HIV, agent-based modeling, health policy, funding impact, Chicago

1. Introduction

Despite advances in HIV prevention and treatment, Black men who have sex with men (MSM) continue to experience disproportionately high rates of HIV infection in the United States [1, 2, 3]. Many US states and jurisdictions have launched initiatives to eliminate HIV over the next decade [2, 3] by expanding biomedical prevention and reducing barriers to engagement and retention in care [4, 5, 6]. These initiatives emphasize approaches tailored to priority populations, including Black MSM, and focus on addressing the social determinants of health that interfere with prevention and care engagement [1, 7, 2]. Illinois launched Getting to Zero Illinois (GTZ-IL) to end HIV in the state by 2030 [8, 9], combining expanded biomedical prevention with efforts to reduce structural and system-level barriers to care.

Recent shifts in US government priorities have raised the prospect of reductions in federal and state HIV program funding, which could reduce access to essential prevention and treatment services. Local policy makers must decide how to use existing programs and infrastructure to sustain progress toward HIV elimination amid changing epidemiological trends and uncertain future funding. Mathematical models can support these decisions by projecting how funding changes might alter HIV transmission across a range of plausible assumptions and future scenarios, particularly valuable when resources are scarce and the future is uncertain. In this study, we examined how reductions in HIV prevention and treatment funding could affect HIV transmission among Black MSM in Chicago. Guided by input from local GTZ-IL stakeholders, we used an agent-based model (ABM) calibrated to the current Chicago epidemic to simulate a range of antiretroviral therapy (ART) and pre-exposure prophylaxis (PrEP) funding scenarios over the next 10 years, and assessed their effects on HIV incidence and prevalence.

2. Methods

2.1. Study Setting

Black MSM, particularly younger men, have been disproportionately impacted by HIV in Chicago since early in the epidemic and represent a key focus population for GTZ-IL. In 2024, Non-Hispanic Black residents accounted for 36% of new HIV diagnoses and 46% of people living with HIV in Chicago despite comprising only 30% of the population, with an incidence rate more

than four times that of non-Hispanic white residents. Although new diagnoses have declined in recent years, young Black MSM have historically had lower rates of PrEP use and engagement in care than other groups. Therefore, this analysis focused on Black MSM aged 16 to 34.

2.2. Model Structure

INFORM-HIV is a stochastic, discrete-time, agent-based network model of HIV transmission among Black MSM aged 16–34 in Chicago, Illinois. Each simulated individual (“agent”) has demographic, behavioral, and clinical attributes that evolve over time. The formation and dissolution of main and casual sexual partnerships are represented as dynamic networks using separable temporal exponential-family random graph models (STERGMS). The probability of HIV transmission within a partnership depends on viral load, disease stage, condom use, and PrEP use, and social and economic factors further shape PrEP use and engagement in HIV care. The model was parameterized to represent the Chicago epidemic using data from 2013 through 2023, reflecting local rates of HIV incidence and prevalence, care engagement, and PrEP use at the end of 2023. Data on sociodemographics, sexual networks, and behavior came from the uConnect study, a longitudinal study of young Black MSM conducted in Chicago between 2013 and 2015.[7, 10, 11] An earlier version of the model has been described previously [12]; further details on design and parameterization are provided in the supplementary appendix.

2.3. Model Calibration

2.3.1. Parameter space reduction via Morris Screening

Many model parameters are uncertain because empirical data are limited or not directly translatable into parameter estimates. We specified plausible ranges for these parameters from empirical data and available literature (Supplementary Table S2). Since a full sensitivity analysis across all uncertain parameters was computationally infeasible, we prioritized parameters that were most uncertain (those with limited or imprecise supporting data) and those expected to most strongly influence outcomes (e.g., transmission probability), yielding 21 parameters for screening. We then used the Morris screening method [13, 14] to reduce dimensionality by identifying the most influential of these 21 inputs, yielding 6 parameters for calibration. Morris screening ranks parameters by their influence on a model output while

accounting for nonlinearities and interactions; full methodological detail is provided in the supplementary appendix.

2.3.2. Calibration to baseline epidemic

After reducing the parameter space to the six most influential parameters, we calibrated the model to estimated HIV prevalence among Black MSM aged 15–34 in Chicago at the end of 2023, the most recent data available at the time. [15] We calibrated the model to HIV prevalence rather than incidence because prevalence tends to be more stable and less sensitive to short-term fluctuations. The baseline (no-change) scenario represents a static snapshot of the epidemic at a single time point, assuming a stationary baseline and a well-mixed population, which is sufficient for the goal of comparing the relative impacts of different strategies for a given baseline. However, more recent trends in incidence may not be captured.

Rather than a single best-fitting parameter set, calibration produced a distribution over plausible parameterizations, from which we retained 100 calibrated model states (“checkpoints”), each reflecting a plausible representation of the epidemic at the end of the calibration period. Propagating this ensemble forward captures uncertainty from the input parameters constrained by the data. When paired with scenario exploration via surrogates (described below), the forward propagation from the checkpoints also enables tracking the surrogate uncertainty along with the parameter uncertainty. The combination of checkpoints and surrogate assisted exploration facilitates a far wider range of scenarios to explore than a full factorial simulation design would allow given the model’s computational cost. Full calibration details are provided in the supplementary appendix.

2.4. Funding Scenarios Elicitation

We defined funding-reduction scenarios in two steps: we first elicited plausible ranges of future ART and PrEP funding changes from local stakeholders (described below), and then mapped those funding changes to changes in population ART and PrEP coverage in the model. (Section 2.5). In July 2025, we consulted a stakeholder group from Getting to Zero Illinois to understand their concerns about HIV funding. Stakeholders (n=7) were asked whether they expected HIV funding to increase, remain stable, or decrease over the next ten years, and what percentage change in ART and PrEP funding they considered plausible relative to current levels. Expected reductions

ranged from 15% to 72% for PrEP and 15% to 68% for ART; no stakeholder expected funding to increase.

2.5. Mapping funding changes to ART and PrEP Use

INFORM-HIV takes as inputs the proportion of seronegative individuals using PrEP and the proportion of seropositive individuals who are virally suppressed on ART. Since the model does not explicitly simulate costs at the individual level, we developed a high-level mapping from program-level expenditures to population-level ART and PrEP coverage among MSM. We denote annual government spending on ART and PrEP by G_{ART} and G_{PrEP} , and the corresponding population coverage levels by P_{ART} and P_{PrEP} .

To allocate national HIV expenditures to the MSM population, we use epidemiological estimates indicating that approximately 52% of HIV-positive individuals in the United States are MSM [5] and that roughly 70% of adults with indications for PrEP are MSM [16]. Using these proportions, baseline annual spending is approximately $G_{\text{ART}} \approx \$18.9$ billion for ART and $G_{\text{PrEP}} \approx \5.7 billion for PrEP (Supplementary Appendix section SA.3).

We make the assumption that coverage increases linearly with government spending, while accounting for two important structural features of the HIV care system. First, some individuals may obtain ART or PrEP from non-federally funded sources (e.g., through private insurance). We represent these groups by the fractions γ_{ART} and γ_{PrEP} , which capture the proportion of MSM whose access is not directly affected by changes in public funding. Second, program delivery is imperfect, and losses can arise from delayed linkage to care, imperfect adherence, and administrative or operational challenges. These effects are captured through scaling factors α for ART and β for PrEP that adjust how efficiently spending translates into coverage.

Putting these components together, coverage is represented as:

$$P_{\text{ART}} = \min\left(1, \frac{G_{\text{ART}}}{\alpha C_{\text{ART}} N_{\text{HIV}+}} + \gamma_{\text{ART}}\right), \quad (1)$$

$$P_{\text{PrEP}} = \min\left(1, \frac{G_{\text{PrEP}}}{\beta C_{\text{PrEP}} N_{\text{HIV-}}} + \gamma_{\text{PrEP}}\right), \quad (2)$$

where $C_{\text{ART}} \approx \$50,000$ and $C_{\text{PrEP}} \approx \$21,000$ denote annual per-person costs (in 2025 USD) [16, 17], and $N_{\text{HIV}+} \approx 638,900$ and $N_{\text{HIV-}} \approx 851,240$ represent

the sizes of the HIV-positive and PrEP-eligible MSM populations, respectively. The parameters α , β , γ_{ART} , and γ_{PrEP} are calibrated so that the model reproduces observed baseline coverage at current funding levels (see Supplementary Appendix). The baseline coverage scenarios for ART and PrEP were 81.50% and 49.11%. Private insurance substitution for government funded coverage resulted in coverage floors of 15.0% and 15.9% for ART and PrEP, respectively.

A proportional funding change is represented by a reduction factor $\Delta \in [0, 1]$, so that government spending becomes $(1 - \Delta)G$. This affects only the publicly financed portion of coverage, while the privately financed component, represented by γ , remains unchanged. This linear structure is chosen for parsimony and interpretability, since the empirical relationship between aggregate funding and coverage is not well-identified at the scale of the reductions considered here. This formulation should be interpreted as a first-order approximation appropriate for moderate funding changes. Its limitations, as well as possible nonlinear extensions, are discussed in the Supplementary Appendix.

Guided by the ranges obtained from stakeholders, we evaluated scenarios reflecting reductions of [0–75]% in ART and PrEP funding, translating each into the corresponding ART and PrEP coverage via Equations (1)–(2).

2.5.1. Simulation of ART and PrEP Scenarios

To evaluate scenarios while accounting for parameter uncertainty, we ran forward simulations from the 100 calibrated model states (i.e., checkpoints). We used each of the 100 checkpoints (Section 2.3) as the initial condition for a 10-year projection under a specified pair of proportional reductions in ART and PrEP coverage, $(r_{\text{ART}}, r_{\text{PrEP}})$. Repeating this across all 100 checkpoints yields, for each scenario, a distribution of projected incidence and prevalence trajectories that reflects uncertainty in the calibrated parameters and epidemic state.

Simulating a fine grid of $(r_{\text{ART}}, r_{\text{PrEP}})$ combinations directly would be computationally prohibitive, since resource use scales with both the number of scenarios and the number of checkpoints. We therefore used surrogate models to approximate these projections efficiently (below).

2.5.2. Surrogate assisted exploration

A surrogate model is a statistical approximation of the simulator that, once trained, predicts outcomes for new intervention settings without rerun-

ning the full model. For each of the 100 checkpoints, we trained Gaussian process surrogates [18] to map a pair of reductions ($r_{\text{ART}}, r_{\text{PrEP}}$) to the projected 10-year incidence and prevalence trajectories (a separate surrogate for each outcome). Training data came from simulating 20 intervention combinations, each with 10 stochastic replicates, at every checkpoint (20 combinations $\times$ 10 replicates $\times$ 100 checkpoints = 20,000 simulations). Because a separate surrogate is trained at each checkpoint, predictions for any intervention combination can be pooled across the 100 checkpoints to give a predictive distribution for 10-year incidence and prevalence that reflects both calibrated parameter uncertainty and the approximation introduced by the surrogates. Further details are provided in the supplementary appendix.

2.6. Outcomes

For each ART and PrEP funding reduction scenario, we summarized projected HIV incidence and prevalence over the 10-year horizon using the surrogate-based predictive distribution pooled across the 100 checkpoints by sampling 50 times from the final year (year 10) from each checkpoint to calculate means with 95% posterior predictive intervals. Scenario comparisons were expressed as rate ratios or absolute differences in mean incidence per 100 person-years relative to the baseline scenario (no change).

2.7. Statistical Analyses

Using the 100 checkpoint-trained surrogates, we predicted incidence and prevalence across a 101-point Latin hypercube design over the coverage-reduction space (0 to -75% for each of ART and PrEP), and performed a linear transformation (Section 2.5) mapping (ART $\times$ PrEP) coverage reductions to combinations of (ART $\times$ PrEP) funding reduction scenarios, by 1% increments respectively. We report 95% posterior predictive intervals for HIV incidence and prevalence in each funding scenario.

3. Results

In the baseline scenario model-projected HIV incidence at 10 years was 4.19 per 100 person-years (95% credible interval [CrI] 3.16–5.06; Table 1). We compared each scenario relative to this baseline and computed mean HIV incidence, the absolute increase in incidence (cases per 100 person-years), and an incidence rate ratio (IRR), each with a 95% CrI. We first report scenarios that reduce antiretroviral therapy (ART) or pre-exposure prophylaxis (PrEP)

use directly (Figure 1 A), then scenarios where we estimate a reduction in ART or PrEP use driven by reductions in government funding (Figure 1 B).

Reducing ART or PrEP use resulted in increases in HIV incidence, with reductions in ART driving the largest increases for a single percentage point use. For example, a 10% reduction in ART use raised incidence to 5.28 per 100 person-years (4.10–6.18), an absolute increase of 1.08 cases per 100 person-years (0.75–1.44) and an IRR of 1.26 (1.18–1.34). A 25% reduction raised incidence to 6.98 per 100 person-years (5.56–8.05; IRR 1.67, 1.54–1.82), and a 40% reduction raised it to 8.58 per 100 person-years (7.11–9.68; IRR 2.06, 1.86–2.28). Reducing PrEP use had a smaller effect at each level: a 10% reduction yielded an IRR of 1.06 (0.99–1.14), a 25% reduction an IRR of 1.16 (1.04–1.31), and a 40% reduction an IRR of 1.25 (1.08–1.48), corresponding to an incidence of 5.26 per 100 person-years (3.89–6.51). Reducing both ART and PrEP use by 40% simultaneously produced the largest impact of any scenario, increasing incidence to 10.12 per 100 person-years (8.46–11.65), an absolute increase of 5.92 cases per 100 person-years (4.88–7.18) and an IRR of 2.43 (2.13–2.78).

Reductions in government funding produced smaller increases in HIV incidence than the equivalent percentage reductions in use, because a funding cut of a given fraction lowers coverage by less than that fraction since not all ART or PrEP use is government-funded (Figure 1). A 10% ART funding reduction mapped to a 7.5% reduction in ART use, raising incidence to 5.00 per 100 person-years (3.85–5.89; IRR 1.19, 1.13–1.26); 25% and 40% ART funding reductions mapped to 18.9% and 30.2% use reductions and IRRs of 1.50 (1.39–1.63) and 1.81 (1.65–1.98), respectively. The dampening was most pronounced for PrEP. A 40% reduction in both ART and PrEP funding raised incidence to 8.34 per 100 person-years (6.85–9.59), an absolute increase of 4.15 cases per 100 person-years (3.49–5.08) and an IRR of 2.00 (1.80–2.25), below the IRR of 2.43 for the equivalent 40% reduction in use.

The impacts of funding reductions were projected to accumulate over time. Their overall impact may therefore be underestimated if examined over shorter time horizons (Figure 2A). In some scenarios, roughly half of the eventual effect of funding reduction emerged within the first two years but those effects continued to grow. For instance, a 40% funding reduction reached an IRR of 1.36 after one year and 1.60 after two, but rose to 1.99 by year 10. Similar trends were observed with smaller funding reductions, reaching year-10 IRRs of 1.23 (10% reduction) and 1.48 (20% reduction). We observed the largest year-10 IRR when funding for ART was cut more than

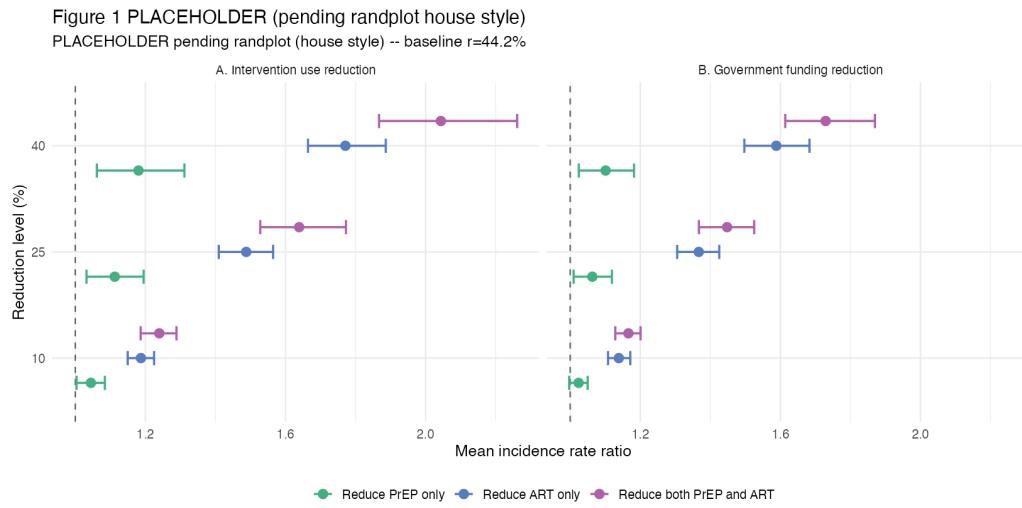


Figure 1: **HIV incidence rises with deeper reductions, but the cost model dampens the effect of funding cuts, especially for PrEP.** Mean HIV incidence rate ratio at 10 years relative to a no-reduction baseline. Panel (A) varies intervention use (coverage) directly; panel (B) varies government funding, which maps to a smaller coverage reduction through the cost model. Within each level, the three points compare reducing PrEP only, ART only, or both equally. Points are posterior means and whiskers are 95% credible intervals, pooled across the 100 calibrated model checkpoints [that are sampled 50 times each]. The dashed line marks the no-change baseline (rate ratio of 1).

Figure 2 PLACEHOLDER (pending randplot house style)
PLACEHOLDER pending randplot (house style) -- baseline $r=44.2\%$

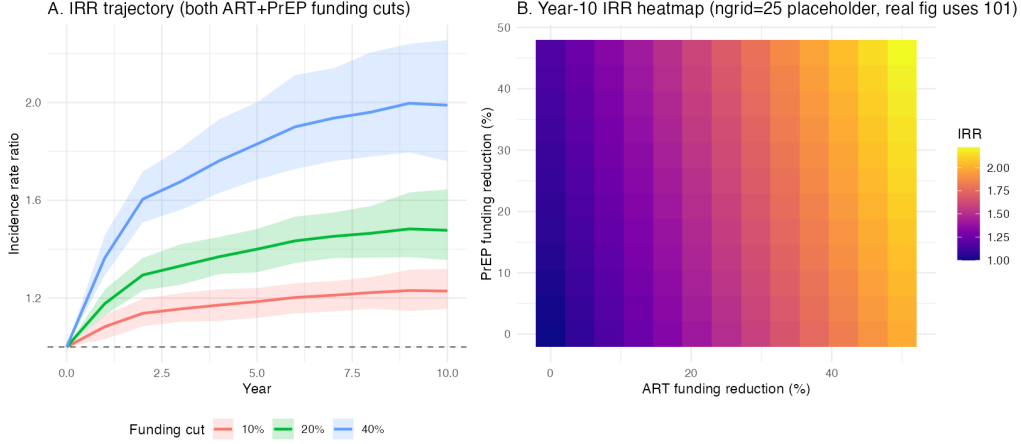


Figure 2: Trajectory of the Impacts of Funding cuts under alternative scenarios. Panel (A) shows the mean HIV incidence rate ratio (relative to a no-reduction baseline) over the 10-year horizon under equal (parallel) reductions to ART and PrEP government funding of 10% (blue), 20% (purple), and 40% (orange). Lines are posterior means and the nested shaded bands are 50% and 95% credible intervals; the y-axis is on a log scale and the dashed line marks the no-change baseline (rate ratio of 1). Panel (B) shows the mean incidence rate ratio at year 10 across all combinations of ART (horizontal) and PrEP (vertical) government funding reductions from 0 to 50%, with color denoting 0.25-wide rate ratio bands. Near-vertical contours indicate that the year-10 rate ratio depends more on the magnitude of the ART funding reduction than on the PrEP reduction. All quantities are pooled across the 100 calibrated model checkpoints.

funding for PrEP (Figure 2B).

Table 1: HIV incidence outcomes under intervention-use and government-funding reductions

Scenario	ART Use Reduction (%)	PrEP Use Reduction (%)	Mean HIV Incidence per 100 p.y. [95% CrI]	Absolute Increase per 100 p.y. [95% CrI]	Incidence Rate Ratio [95% CrI]
Baseline (no cuts)	0.0	0.0	4.19 [3.16, 5.06]	0.00 [0.00, 0.00]	1.00 [1.00, 1.00]
A. Intervention use reduction					
10% ART cut only	10.0	0.0	5.28 [4.10, 6.18]	1.08 [0.75, 1.44]	1.26 [1.18, 1.34]
25% ART cut only	25.0	0.0	6.98 [5.56, 8.05]	2.78 [2.27, 3.31]	1.67 [1.54, 1.82]
40% ART cut only	40.0	0.0	8.58 [7.11, 9.68]	4.38 [3.82, 5.05]	2.06 [1.86, 2.28]
10% PrEP cut only	0.0	10.0	4.46 [3.34, 5.36]	0.26 [-0.02, 0.57]	1.06 [0.99, 1.14]
25% PrEP cut only	0.0	25.0	4.86 [3.63, 5.86]	0.66 [0.17, 1.21]	1.16 [1.04, 1.31]
40% PrEP cut only	0.0	40.0	5.26 [3.89, 6.51]	1.06 [0.33, 1.90]	1.25 [1.08, 1.48]
40% cut for both	40.0	40.0	10.12 [8.46, 11.65]	5.92 [4.88, 7.18]	2.43 [2.13, 2.78]
B. Government funding reduction					
10% ART cut only	7.5	0.0	5.00 [3.85, 5.89]	0.80 [0.52, 1.11]	1.19 [1.13, 1.26]
25% ART cut only	18.9	0.0	6.29 [4.96, 7.29]	2.09 [1.65, 2.55]	1.50 [1.39, 1.63]
40% ART cut only	30.2	0.0	7.54 [6.08, 8.66]	3.35 [2.81, 3.92]	1.81 [1.65, 1.98]
10% PrEP cut only	0.0	5.6	4.34 [3.26, 5.23]	0.15 [-0.07, 0.36]	1.03 [0.98, 1.09]
25% PrEP cut only	0.0	14.0	4.56 [3.43, 5.47]	0.37 [0.04, 0.79]	1.09 [1.01, 1.19]
40% PrEP cut only	0.0	22.3	4.79 [3.58, 5.75]	0.59 [0.13, 1.12]	1.14 [1.03, 1.27]
40% cut for both	30.2	22.3	8.34 [6.85, 9.59]	4.15 [3.49, 5.08]	2.00 [1.80, 2.25]

Abbreviations: ART, antiretroviral therapy; PrEP, pre-exposure prophylaxis; HIV, human immunodeficiency virus; p.y., person-years; CrI, credible interval; IRR, incidence rate ratio.

All quantities are HIV incidence outcomes at the 10-year horizon, reported as the mean with a 95% CrI pooled across the 100 calibrated model checkpoints. *ART Use Reduction* and *PrEP Use Reduction* are the reductions in intervention use (coverage) fed to the model: in block A they equal the labelled reduction, whereas in block B they are the smaller reductions in use implied by the labelled *funding* reduction through the cost model, because a funding cut of a given fraction lowers coverage by less than that fraction (the effect is smallest for PrEP, a larger share of whose access is privately financed). *Mean HIV Incidence* is new HIV infections per 100 person-years. *Absolute Increase* is the difference in incidence from the baseline (no-cut) scenario, in cases per 100 person-years. *Incidence Rate Ratio* is the ratio of scenario incidence to baseline incidence; a value of 1 indicates no change. Both blocks share the single baseline row.

4. Discussion

Our findings indicate that reductions in HIV program funding could substantially increase HIV transmission among Black MSM in Chicago. Even modest reductions in ART resulted in meaningful increases in transmission over 10 years. A 10% reduction in ART resulted in a 26% increase in HIV incidence relative to the baseline scenario, while a 25% reduction in ART resulted in a 67% increase in HIV incidence in the tenth year. In a population of 5,000 Black MSM, this translates to 54 and 139 new infections per year, respectively. Reductions in PrEP use also resulted in increased incidence, though to a lesser extent than ART, which is likely due at least in part to assumptions about PrEP adherence and delivery strategies. The impact of PrEP reductions would be expected to be smaller when ART coverage is maintained, and to become increasingly important in scenarios where incidence is high or viral suppression is low. These results underscore the critical role of sustained program funding in preventing HIV transmission.

Our findings are consistent with recent studies have demonstrated the potential impact at the national level of reductions in PrEP [19] and Ryan White program funding [20].

Reductions in federal funding could result in increased disparities as those who rely on publicly funded programs for medication will be most heavily impacted.[cite] The drivers of ongoing HIV transmission are complex and rooted in social and economic inequity. HIV is increasingly concentrated in communities with the fewest resources and the least access to prevention and care. As biomedical interventions have reduced HIV in some subgroups, elimination will increasingly depend on reaching smaller, higher-need segments of the population, an effort that becomes more difficult and costly as that population grows. Sustained funding for biomedical prevention and treatment is therefore necessary but not sufficient: progress also depends on addressing housing, mental health, and substance use, all of which could be affected by future funding decisions. Reductions in funding for these supportive services, alongside direct cuts to HIV programs, could act synergistically to accelerate transmission.

Short-term cost-savings due to reductions in HIV prevention funding could be offset by increased healthcare costs long-term. It may take a long time to get back to where we were in terms of downward trajectories if essential program infrastructure is removed.

Increases in new HIV diagnoses have occurred in Chicago [21] and other

areas (cites) and funding reductions could exacerbate or accelerate these trends.

4.1. Limitations

Several limitations should be considered. We assumed a simple linear mapping between funding reductions and changes in ART and PrEP use. In reality this relationship is likely more complex, as funds may be shifted from other programs to ensure access to HIV treatment for as long as possible. We also did not model reductions in funding for social services such as housing, which could accompany cuts to direct HIV services. For these reasons, the overall impact on the epidemic may be underestimated.

The relative effects of reductions in ART and PrEP depend on baseline coverage and adherence. Because baseline PrEP and ART coverage differ, an identical percentage reduction does not translate into the same absolute change in the number of people using each. ART coverage is also comparatively high at baseline, so reducing it by a given percentage has a larger effect than the same percentage reduction for PrEP. Direct comparisons of the relative impact of ART versus PrEP reductions should therefore be made with caution; the results are best interpreted as a range of plausible outcomes for a given funding reduction, depending on how cuts are distributed across programs.

Finally, our findings may not generalize to other populations or settings, and could be more or less optimistic depending on the magnitude of funding cuts and the availability of alternative funding.

Overall rates of reliance on government funding are likely underestimated for Black MSM.

4.2. Conclusions

Future work should examine strategies to mitigate the effects of funding reductions and to use limited resources most efficiently by prioritizing the highest-impact interventions and implementation strategies.

Supplementary Appendix

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The design, parameterization, and processes for the Integrated Framework for Modeling Social Determinants of HIV (INFORM-HIV) agent-based model have been previously published. [cite preprint] This Supplementary Appendix describes the calibration process, experimental design, and methods for mapping reductions in HIV program funding to reductions in ART and PrEP use.

SA.1. Model Calibration

SA.1.1. Calibration targets

Direct estimates of HIV incidence among young Black MSM are not available from routine surveillance, and it is difficult to establish which estimates best approximate true incidence: surveillance data are subject to reporting delays, misclassification of risk behavior and gender identity, and undercounting of undiagnosed infections, while cohort estimates may be imprecise or non-representative. For reference, HIV incidence in the uConnect cohort was 8.5 per 100 person-years (95% CI 6.0–11.9; 33 incident infections among 300 participants confirmed HIV-negative at baseline).[22] Prevalence estimates share some of these limitations but are more stable over time and more reliably estimated from population-level surveillance. We therefore used HIV prevalence from surveillance data as the primary calibration target – specifically, the mean prevalence over the final five simulated years, with a target value of 29.4% (Table S1).

SA.1.2. Morris Screening

Due to the limitations of existing data, the values for multiple model parameters were uncertain. We explored plausible ranges for these parameters based on empirical data and existing literature as part of the calibration process to select the most likely set of parameter values that approximated the target HIV prevalence. Because of the large number of parameters, we were not able to conduct full sensitivity analysis on all of them. Therefore we selected the most uncertain parameters (i.e., those for which there was limited data or existing data was considered imprecise) and those considered to have a large impact on model outcomes (eg transmission probability), which resulted in 21 parameters for Morris screening.

Before model calibration, we performed a dimension reduction using the Morris screening method [13, 14] to identify the most influential parameters

Table S1: HIV prevalence estimates among Black MSM in Chicago used as calibration targets.

Group / period	Estimate	Source
Approximate number of Black MSM ages 13-34 living with HIV as of 2023	1723	[15]
Approximate population size of Black males ages 15-34 in Chicago (2021 ACS)	116853	[23]
Approximate number of Black MSM ages 15-34 in Chicago	6544	[24, 23]
Undiagnosed infection among young Black MSM in Chicago	12%	[25]
Approximate HIV prevalence (%) among young Black MSM in Chicago	29.49	Derived from above

among the 21 uncertain model inputs. In the Morris screening method, elementary effects are computed, and the model inputs are ranked according to their influence on model outputs while accounting for potential nonlinearities and interactions. Briefly, the approach perturbs each parameter one at a time across a discretized grid of the input space, while randomly sampling the remaining parameters within their respective ranges. For each parameter x_i , the corresponding elementary effect is computed as

$$EE_i = \frac{Y(\mathbf{x}_1, \dots, x_i + \Delta, \dots, \mathbf{x}_k) - Y(\mathbf{x})}{\Delta}, \quad (\text{A.1})$$

where $Y(\cdot)$ denotes the model output of interest (in our case, the average HIV prevalence), and Δ is a predetermined step size within the normalized $[0, 1]$ input space. Repeating this process over multiple random trajectories yields a distribution of elementary effects for each parameter. Two summary statistics are typically derived: (i) the mean of the absolute elementary effects, μ_i^* , which reflects the overall influence of parameter i on the output, and (ii) the standard deviation, σ_i , which captures nonlinearity and interaction effects. Parameters with large μ_i^* are considered highly influential, whereas those with large σ_i indicate complex effects or interactions.

We implemented the Morris design using $r = 20$ trajectories and $p = 6$ grid levels. Based on the joint examination of μ_i^* and σ_i values, six parameters were identified as key drivers of the model outputs and retained for

subsequent calibration. The selected parameters and their ranges are summarized in Table S2, while Figure S1 shows the corresponding sensitivity indices.

Table S2: Model parameters and ranges used in the Morris screening analysis.

Symbol	Parameter	Description	lb	ub
θ_1	prep.bl.use.prop	Proportion using PrEP	1.00×10^{-1}	4.00×10^{-1}
θ_2	prep.mean.days.usage.0	Mean duration of PrEP use (days)	6.00×10^1	5.40×10^2
θ_3	art.init.lag.mean	Time from diagnosis to ART initiation (days)	7.00	2.70×10^2
θ_4	non.testers.prop.0	Proportion who never test for HIV	5.00×10^{-2}	1.50×10^{-1}
θ_5	tests.per.year.0	If ever tested, no. of tests per year	5.00×10^{-1}	4.00
θ_6	transmission.probability.insertive.infected	Per act transmission probability if insertive partner is infected	4.79×10^{-4}	1.04×10^{-3}
θ_7	transmission.probability.receptive.infected	Per act transmission probability if receptive partner is infected	5.32×10^{-5}	4.15×10^{-4}
θ_8	seroconcordant.main.condom.prob	Probability of condom use with seroconcordant main partners	5.00×10^{-2}	5.90×10^{-1}
θ_9	serodiscordant.main.condom.prob	Probability of condom use with serodiscordant main partners	2.00×10^{-1}	7.50×10^{-1}
θ_{10}	seroconcordant.casual.condom.prob	Probability of condom use with seroconcordant casual partners	2.00×10^{-1}	7.90×10^{-1}
θ_{11}	serodiscordant.casual.condom.prob	Probability of condom use with serodiscordant casual partners	2.00×10^{-1}	8.30×10^{-1}
θ_{12}	prop.steady.sex.acts	Frequency of sex in main partnerships (acts/week)	1.20×10^{-1}	2.90×10^{-1}
θ_{13}	prop.casual.sex.acts	Frequency of sex in casual partnerships (acts/week)	7.00×10^{-2}	1.40×10^{-1}
θ_{14}	incarceration.prob	Daily probability of incarceration with no prior incarceration history	3.85×10^{-5}	1.54×10^{-4}
θ_{15}	incarceration.with.cji.prob	Daily probability of incarceration if previously incarcerated	3.85×10^{-4}	8.56×10^{-4}
θ_{16}	jail.serving.time.mean	Mean duration of incarceration (days)	2.00	1.80×10^2
θ_{17}	network.retention.multiplier	Relative probability of continuing partnership post-jail	2.00×10^{-1}	1.00
θ_{18}	post.release.interference.period.mean	Duration of disrupted ART/PrEP access post-release (days)	3.00×10^1	3.65×10^2
θ_{19}	substance.use.condom.prob.multiplier	Relative probability of condom use if either partner uses substances	2.00×10^{-1}	1.00
θ_{20}	external.infections.per.person.day.min	Lower bound for probability of external infection	3.84×10^{-6}	1.84×10^{-5}
θ_{21}	external.infections.per.person.day.max	Upper bound for probability of external infection	6.85×10^{-6}	3.29×10^{-5}

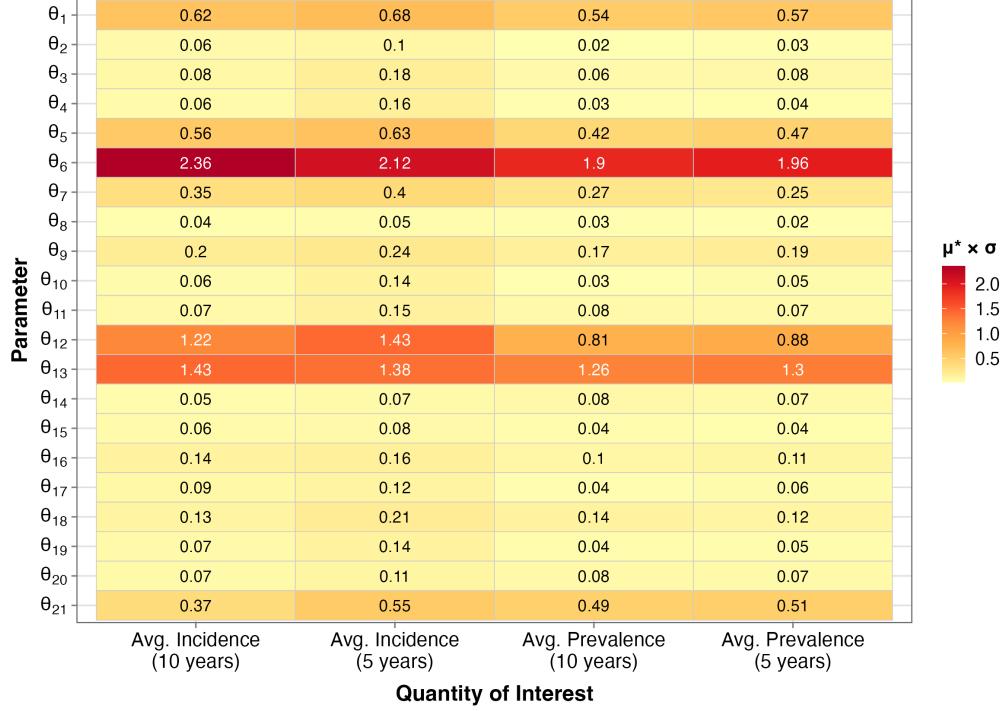


Figure S1: Morris screening: combined sensitivity index $\mu^* \times \sigma$ for each parameter θ_i (Table S1) across four outputs (average HIV incidence and prevalence, 5- and 10-year windows). Darker cells denote more influential parameters.

SA.1.3. Bayesian Calibration

Following Morris screening, the reduced six parameter INFORM-HIV model was calibrated to reproduce the observed HIV prevalence. Let $\boldsymbol{\theta} \in \Theta \subset \mathbb{R}^6$ denote the retained parameter vector. For each parameter setting, the stochastic simulator was evaluated using multiple independent random seeds to account for intrinsic simulation variability. Annual HIV prevalence was recorded from each simulation, and the calibration summary statistic was defined as the average prevalence over the final five simulated years. The observed calibration target was the corresponding empirical five-year average HIV prevalence, $y_{\text{obs}} = 0.294$.

Calibration was performed using simulation-based inference [26] with sequential neural posterior estimation (NPE). A prior distribution was specified over the reduced parameter space, and the simulator was treated as an

implicit generative model that maps parameter values to the prevalence summary statistic through forward simulation. NPE learns an approximation to the posterior distribution $\pi(\boldsymbol{\theta} \mid y_{\text{obs}})$, which characterizes parameter values consistent with the observed prevalence.

Training data were generated using a Latin hypercube design with $N = 100$ parameter settings over the reduced parameter space. At each design point, the INFORM-HIV model was simulated $R = 10$ times using independent random seeds, resulting in 1,000 simulations. The same set of random seeds was used across all parameter settings to reduce Monte Carlo variability when comparing simulator outputs. The resulting parameter-summary pairs were used to train the sequential NPE model. After training, 100 parameter vectors were sampled from the approximate posterior and used as calibrated parameter sets for the subsequent scenario analysis.

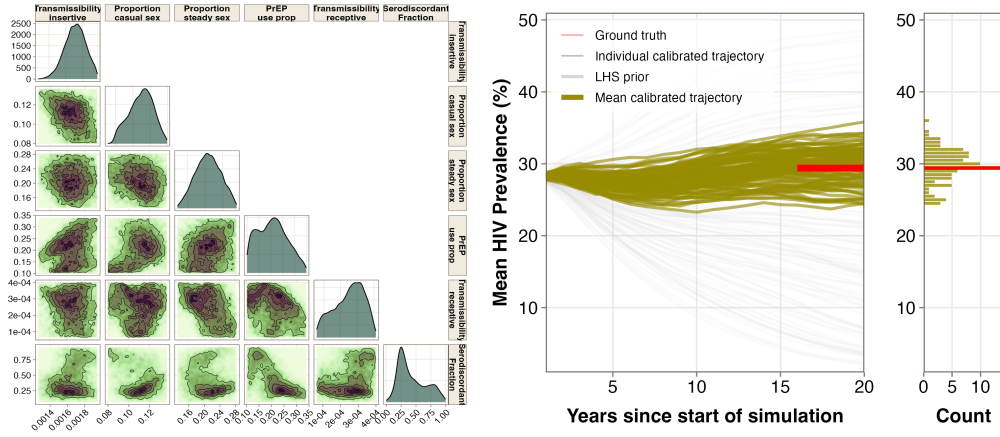


Figure S2: Calibration of the INFORM-HIV model. Left: posterior marginal (diagonal) and pairwise joint (lower triangle) densities of the six calibrated parameters. Right: posterior-predictive HIV prevalence trajectories (LHS prior in faint grey, and posterior means are highlighted by yellow) with the final-year prevalence histogram; the red line marks the 29.4% calibration target.

SA.2. ART and PrEP Reduction Scenario Exploration

Following calibration, 100 posterior samples were retained as checkpointed model states at the end of the 20-year calibration period. Each checkpoint was propagated forward for an additional 10 years under intervention settings defined by percent reductions in ART coverage and PrEP use,

$\mathbf{u} = (r_{\text{ART}}, r_{\text{PREP}}) \in \mathcal{U} \subset \mathbb{R}^2$, relative to the baseline. Direct evaluation over a dense intervention grid is computationally expensive because every intervention must be simulated for each checkpointed model state.

To accelerate scenario exploration, a Gaussian process (GP) surrogate [18] was constructed for each checkpoint. For a fixed checkpoint, let $\mathbf{y}(\mathbf{u}) \in \mathbb{R}^T$ denote the projected prevalence or incidence trajectory over the next $T = 10$ years. The trajectory was represented using a low-dimensional basis expansion

$$\mathbf{y}(\mathbf{u}) \approx \Phi \mathbf{w}(\mathbf{u}),$$

where $\Phi \in \mathbb{R}^{T \times J}$ contains basis vectors obtained from singular value decomposition and $\mathbf{w}(\mathbf{u}) \in \mathbb{R}^J$ denotes the corresponding basis coefficients. Independent GP models were fitted to each coefficient, yielding the predictive distribution

$$\mathbf{y}(\mathbf{u}) \mid \boldsymbol{\theta}^{(k)} \sim N(\boldsymbol{\mu}_k(\mathbf{u}), \boldsymbol{\Sigma}_k(\mathbf{u})),$$

where $\boldsymbol{\theta}^{(k)}$ denotes the k th posterior sample retained during calibration.

Training data for each surrogate consisted of an initial Latin hypercube design of 20 intervention combinations with 10 stochastic replicates per combination. Separate surrogate models were constructed for prevalence and incidence at each checkpoint.

The final predictive distribution was obtained by averaging over the posterior samples retained during calibration,

$$p(\mathbf{y}(\mathbf{u}) \mid y_{\text{obs}}) \approx \frac{1}{K} \sum_{k=1}^K N(\boldsymbol{\mu}_k(\mathbf{u}), \boldsymbol{\Sigma}_k(\mathbf{u})),$$

where $K = 100$, and $\boldsymbol{\mu}_k(\cdot), \boldsymbol{\Sigma}_k(\cdot)$ are the mean and covariance of the fitted GP surrogate. This multi-surrogate formulation propagates uncertainty from both Bayesian calibration and surrogate prediction into the projected intervention outcomes.

SA.3. Mapping Funding Reductions to PrEP and ART use

This section describes the derivation of the expenditure estimates and parameter values used to map changes in U.S. government HIV funding to changes in ART and PrEP coverage. The main paper presents the coverage equations; here we provide full description of all inputs to those equations.

SA.3.1. Model Notation

We model the relationship between annual U.S. government investment in HIV prevention and treatment and the resulting population-level coverage of ART and PrEP among MSM. The key quantities are:

- P_{ART} : proportion of HIV-positive MSM receiving ART
- P_{PrEP} : proportion of PrEP-eligible HIV-negative MSM using PrEP
- G_{ART} : annual government spending on ART allocated to MSM
- G_{PrEP} : annual government spending on PrEP allocated to MSM
- C_{ART} : annual per-person cost of ART (2025 USD)
- C_{PrEP} : annual per-person cost of PrEP (2025 USD)
- $N_{\text{HIV}+}$: number of HIV-positive MSM in the United States
- $N_{\text{HIV}-}$: number of PrEP-eligible HIV-negative MSM in the United States
- α : ART program inefficiency factor (accounts for delayed linkage, sub-optimal adherence, and administrative overhead)
- β : PrEP program inefficiency factor (accounts for drop-off and adherence losses)
- γ_{ART} : proportion of HIV-positive MSM accessing ART through non-governmental sources (e.g., private insurance)
- γ_{PrEP} : proportion of PrEP-eligible MSM accessing PrEP through non-governmental sources

SA.3.2. U.S. Government Expenditure Estimates

Total federal spending on HIV care and treatment in FY 2022 was approximately \$28.7 billion (base) plus \$6.7 billion (discretionary) = \$35.4 billion for ART-related programs, and \$7.1 billion (base) plus \$1.0 billion (discretionary) = \$8.1 billion for PrEP-related programs [27].

To allocate these national figures to the MSM population, we use two epidemiological proportions:

- MSM account for approximately 52.4% of HIV-positive individuals in the United States: 638,900 MSM out of 1,218,400 total HIV-positive individuals (2022) [5].
- MSM account for approximately 70.0% of adults with PrEP indications in the United States: 851,240 MSM out of 1,216,210 eligible adults (2018) [16].

These proportions are national U.S. estimates covering all MSM regardless of race/ethnicity and all adult age groups.

Applying these proportions to the total spending figures:

$$G_{\text{ART}} = \$35.4\text{B} \times 0.524 \approx \$18.9 \text{ billion}, \quad (\text{C.1})$$

$$G_{\text{PrEP}} = \$8.1\text{B} \times 0.700 \approx \$5.7 \text{ billion}. \quad (\text{C.2})$$

SA.3.3. Per-Person Cost Parameters

ART cost (C_{ART}). The annual per-person cost of ART for a virally suppressed individual was set to $C_{\text{ART}} = \$50,000$ (2025 USD), taken as the midpoint of a payer-stratified range derived from Bingham et al. 2021 [17]. That paper reports monthly cost parameters for HIV-positive adults with $\text{CD4} > 500 \text{ cells}/\mu\text{L}$ (virally suppressed stratum), which annualise to \$36,227/year (2019 USD); inflating by the BLS CPI-U ratio (2019–2025: $\times 1.259$) yields \$45,600/year as a government-payer lower bound. A WAC-price upper bound of approximately \$52,000/year is consistent with 2023–2024 list prices for Biktarvy plus standard outpatient and laboratory costs. We use \$50,000 as the central base-case estimate and explore \$45,600–\$54,000 in sensitivity analysis.

PrEP cost (C_{PrEP}). The annual per-person cost of PrEP was set to $C_{PrEP} = \$21,000$ (2025 USD), representing the total per-person cost of delivering PrEP (medication plus clinical visits plus laboratory monitoring) averaged across insurance types. This figure reflects a weighted average of brand-name (Truvada/Descovy, WAC $\approx \$22,000$ – $\$24,000$ /year) and generic TDF/FTC (340B/FSS prices, $\approx \$8,000$ – $\$13,000$ /year) medication costs combined with annual clinical and laboratory costs of $\$1,000$ – $\$2,000$, weighted by the insurance mix of current PrEP users (approximately 78% private, 22% Medicaid/uninsured) [16].

SA.3.4. Population Size Parameters

HIV-positive MSM in the U.S. were estimated at $N_{HIV+} = 638,900$ based on 2022 CDC surveillance data [5]. HIV-negative MSM with PrEP clinical indications were estimated at $N_{HIV-} = 851,240$ based on 2018 PrEP indication estimates [16]. Both figures are summarized in Table S3.

Table S3: Population parameters used in the funding-to-coverage model

Parameter	Value	Source
N_{HIV+} : HIV-positive MSM in U.S. (2022)	638,900	[5]
Total HIV-positive in U.S. (2022)	1,218,400	[5]
N_{HIV-} : MSM with PrEP indications (2018)	851,240	[16]
Total adults with PrEP indications (2018)	1,216,210	[16]

SA.3.5. Calibration of Inefficiency and Substitution Parameters

At the observed baseline (current funding and coverage), the inefficiency parameters α and β are solved from the coverage equations so that predicted coverage matches empirical coverage:

$$\alpha = \frac{G_{ART}}{(P_{ART} - \gamma_{ART}) C_{ART} N_{HIV+}}, \quad (C.3)$$

$$\beta = \frac{G_{PrEP}}{(P_{PrEP} - \gamma_{PrEP}) C_{PrEP} N_{HIV-}}. \quad (C.4)$$

The private-substitution floors γ_{ART} and γ_{PrEP} represent the shares of MSM who retain access to ART or PrEP through private insurance or other

means when government funding is reduced. These are informed by data on insurance coverage patterns among MSM: roughly 20–25% of HIV-positive individuals receive care through Ryan White/ADAP, with the remainder covered by Medicaid, Medicare, or private insurance [27], and a large share of PrEP prescriptions are reimbursed via private insurance [28, 6]. Adherence and persistence data from CDC surveillance and observational studies indicate material losses between initiation and sustained engagement for both ART and PrEP [29].

Table S4 summarizes the resulting parameter estimates used in the base-case analysis.

Table S4: Estimated parameter values for the ART and PrEP coverage model

Parameter	Description	Range	Base case
α	ART inefficiency factor	1.2–1.4	1.26
β	PrEP inefficiency factor	3.0–5.0	1.58
γ_{ART}	Private ART access fraction	0.10–0.25	0.15
γ_{PrEP}	Private PrEP access fraction	0.20–0.35	0.16

Base-case values were set as $\alpha = 1.26$, $\beta = 1.58$, $\gamma_{\text{ART}} = 0.15$, and $\gamma_{\text{PrEP}} = 0.16$ (Table S4).

The $\gamma_{\text{ART}} = 0.15$ value reflects the approximately 15% of HIV-positive MSM who access ART via private insurance; the remaining 85% are covered by Medicaid, Medicare, or Ryan White CARE Act funds [27]. The $\gamma_{\text{PrEP}} = 0.16$ value is derived as $0.44 \times P_{\text{PrEP}}^0$, the privately insured share of current PrEP users multiplied by the observed baseline PrEP coverage [6, 16]. The inefficiency factors are consistent with empirical estimates of linkage-to-care rates and adherence patterns stratified by payer source [29].

The large calibrated value $\beta \approx 4.0$ reflects the fact that G_{PrEP} encompasses broad HIV prevention infrastructure costs (outreach, navigation, clinic support) beyond medication reimbursement; it should be interpreted as a program overhead factor rather than a per-unit service inefficiency. This is acknowledged as a limitation of the current linear model.

SA.3.6. Coverage Predictions Under Funding Reductions

Let $\Delta \in [0, 1]$ denote a proportional funding reduction, so that government spending under the reduced scenario is $(1 - \Delta)G$. Coverage under a funding reduction is then:

$$P_{\text{ART}}(\Delta) = \min\left(1, \frac{(1 - \Delta) G_{\text{ART}}}{\alpha C_{\text{ART}} N_{\text{HIV}+}} + \gamma_{\text{ART}}\right), \quad (\text{C.5})$$

$$P_{\text{PrEP}}(\Delta) = \min\left(1, \frac{(1 - \Delta) G_{\text{PrEP}}}{\beta C_{\text{PrEP}} N_{\text{HIV-}}} + \gamma_{\text{PrEP}}\right). \quad (\text{C.6})$$

The privately financed component γ is unaffected by Δ ; only the publicly financed share of coverage declines. The $\min(1, \cdot)$ cap ensures that coverage cannot exceed 100%.

For example, a 50% reduction in government ART spending reduces ART coverage by less than 50%; the actual decline is $(1 - \gamma_{\text{ART}}) \times 50\% \approx 42.5$ percentage points starting from baseline, before the min constraint is applied.

SA.3.7. Limitations of the Linear Approximation and Directions for Future Work

The main paper adopts a linear formulation for the mapping from government spending to ART and PrEP coverage, chosen for parsimony and because the true empirical relationship between aggregate funding and coverage is not well-identified at the scale of the reductions considered. This section elaborates on the sources of nonlinearity that the linear model abstracts away, and outlines how future work could incorporate them.

Why the linear model is a first-order approximation.. In practice, the true response of coverage to funding changes is likely nonlinear. Large funding cuts may induce threshold effects such as clinic closures, workforce contraction, or supply-chain disruptions, which would amplify reductions in coverage beyond what a proportional model predicts. Moreover, the impact of funding changes is heterogeneous across insurance and socioeconomic strata: privately insured individuals are largely insulated from short-term changes in public funding, while uninsured populations — who depend most heavily on Ryan White/ADAP and Medicaid — are more directly and severely affected. Neither of these mechanisms is captured by the constant-parameter linear model.

Three nonlinear mechanisms.. Three concrete sources of nonlinearity are abstracted away in the current model:

1. **Insurance heterogeneity.** MSM are stratified by insurance status (private, Medicaid, uninsured). A uniform percentage reduction in government spending produces highly unequal impacts across strata: privately insured individuals are largely insulated (high private substitution floor γ), while uninsured individuals face steep declines (low γ , high α/β). The aggregate population-level response is therefore nonlinear even though each stratum responds linearly to its allocated funding. An illustrative calculation suggests that a 30% budget cut might reduce ART coverage among the privately insured by only 2–3 percentage points, Medicaid enrollees by 15–20 points, and the uninsured by 40–50 points — yielding an aggregate decline substantially larger than the naive linear prediction.
2. **Threshold and infrastructure collapse.** Healthcare delivery systems carry fixed costs (staffing, facilities, equipment). Sufficiently large funding cuts can trigger clinic closures or workforce losses that amplify coverage declines beyond linear predictions once spending falls below a structural threshold (estimated at roughly 65–75% of baseline in related literature). Below this threshold, the effective inefficiency of the remaining system rises sharply as economies of scale are lost.
3. **Dynamic inefficiency.** The parameters α and β are held constant across all funding levels in the current model. In reality, large cuts are likely to increase inefficiency (loss of economies of scale, higher administrative overhead per patient), while funding expansions may improve it. Allowing α and β to vary with spending level would capture this feedback.

Implications for the current analysis.. For the funding reductions considered in the main analysis (up to 75% of baseline), the linear model likely understates coverage losses among uninsured and publicly insured MSM while providing a reasonable approximation for privately insured subpopulations. Overall population-level estimates should therefore be interpreted as conservative: true coverage losses could be larger if threshold effects or insurance-heterogeneity mechanisms are operative. Sensitivity of results to the choice of γ , α , and β can be partially assessed by varying these parameters within the ranges reported in Table S4. Table S5 reports this directly for γ_{PrEP} 's underlying payer-mix parameter r at $r = 25\%$ (unvalidated placeholder pending AFC), $r = 44.2\%$ (base case), and $r = 78\%$ (prior national estimate); $P^0 = 35.9\%$ is not varied separately, since $(P^0 - \gamma_{\text{PrEP}})/P^0 = 1 - r$ is alge-

braically independent of P^0 .

Table S5: Sensitivity of the incidence rate ratio (mean [95% CrI]) to the PrEP payer-mix parameter r , at lower/baseline/upper point values.

Scenario	Funding cut	Lower ($r = 25\%$)	Baseline ($r = 44.2\%$)	Upper ($r = 78\%$)
Reduce PrEP only	10%	1.03 [1.00, 1.06]	1.02 [1.00, 1.05]	1.01 [1.00, 1.02]
	25%	1.09 [1.02, 1.16]	1.06 [1.01, 1.12]	1.02 [1.00, 1.05]
	40%	1.13 [1.04, 1.23]	1.10 [1.03, 1.18]	1.04 [1.00, 1.08]
Reduce ART only	10%	1.14 [1.11, 1.17]	1.14 [1.11, 1.17]	1.14 [1.11, 1.17]
	25%	1.37 [1.30, 1.42]	1.37 [1.30, 1.42]	1.37 [1.30, 1.42]
	40%	1.59 [1.50, 1.68]	1.59 [1.50, 1.68]	1.59 [1.50, 1.68]
Reduce both PrEP and ART	10%	1.18 [1.13, 1.21]	1.17 [1.13, 1.20]	1.15 [1.12, 1.18]
	25%	1.47 [1.38, 1.57]	1.45 [1.37, 1.52]	1.40 [1.34, 1.46]
	40%	1.78 [1.64, 1.94]	1.73 [1.61, 1.87]	1.65 [1.56, 1.76]

Future work. Incorporating insurance stratification is the most tractable near-term extension: data on population shares by insurance status, stratum-specific substitution rates, and stratum-specific inefficiency factors are available from CDC National HIV Behavioral Surveillance, Ryan White Program Services Reports, and Medicaid enrollment records. Threshold and dynamic-inefficiency effects would require additional data on provider-level fixed costs and historical responses to funding shocks (e.g., sequestration impacts on Ryan White, state-level ADAP waiting lists). We defer these extensions to future work.

SA.4. Software

The model was implemented in Python using the Repast for Python (Repast4Py) high-performance agent-based modeling toolkit [30]. Exponential-family random graph models were fit and simulated using the `statnet` suite of R packages [31, 32], interfaced from Python via `rpy2`. Simulations were run on the Midway3 high-performance computing cluster at the University of Chicago. Code to reproduce the experiments and analyses is available at [https://github.com/\[your-org\]/\[repo\]](https://github.com/[your-org]/[repo]).

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